Software and Hardware Tools

1. Install Wireshark on your laptop and capture 2 minutes of network traffic while you browse Instagram or YouTube; save the capture as a .pcap file and note the top 3 protocols you see.

TLS & QUIC (Web & Media Streaming)

- **TLS over TCP:** Secures standard HTTP/1.1 and HTTP/2 connections. Uses a TCP 3-way handshake followed by a TLS handshake. Data payloads are encrypted, hiding application content like video streams or web data from plain inspection.
- **QUIC over UDP:** Underpins HTTP/3. Combines transport and encryption into a single handshake directly over UDP port 443. Eliminates head-of-line blocking by running multiple independent streams over a single connection.

Frame 14: 1250 bytes on wire
UDP, Src Port: 443, Dst Port: 54122
QUIC
  [QUIC Connection Information]
  Header Form: Long Packet Type: Initial
  Version: 1 (0x00000001)
  Destination Connection ID: 0x8f2d91a...
  TLSv1.3 Record Layer: Handshake Protocol: Client Hello
    Cipher Suites (17 suites)
    Extension: server_name (len=18)
      Server Name: googlevideo.com

SSDP — Simple Service Discovery Protocol

- **Layer/Port:** UDP / Port 1900.
- **Multicast Address:** `239.255.255.250` (IPv4).
- **How It Works:** Uses HTTPU (HTTP over UDP) headers like `M-SEARCH` and `NOTIFY`. Smart TVs, media servers, and Universal Plug and Play (UPnP) devices use it to announce their presence and capabilities to local clients.

NOTIFY * HTTP/1.1
HOST: 239.255.255.250:1900
CACHE-CONTROL: max-age=1800
LOCATION: http://192.168.1.45:8008/ssdp/device-desc.xml
SERVER: Linux/3.10 UPnP/1.0 UPnP-Device-Host/1.0
NT: urn:dial-multiscreen-org:service:dial:1
USN: uuid:1e23f982-3200-1000-8000-a47e39a039b0::urn:dial-multiscreen-org:service:dial:1

mDNS — Multicast DNS

- **Layer/Port:** UDP / Port 5353.
- **Multicast Address:** `224.0.0.251` (IPv4) or `ff02::fb` (IPv6).
- **How It Works:** Resolves hostnames ending in `.local` without requiring a centralized

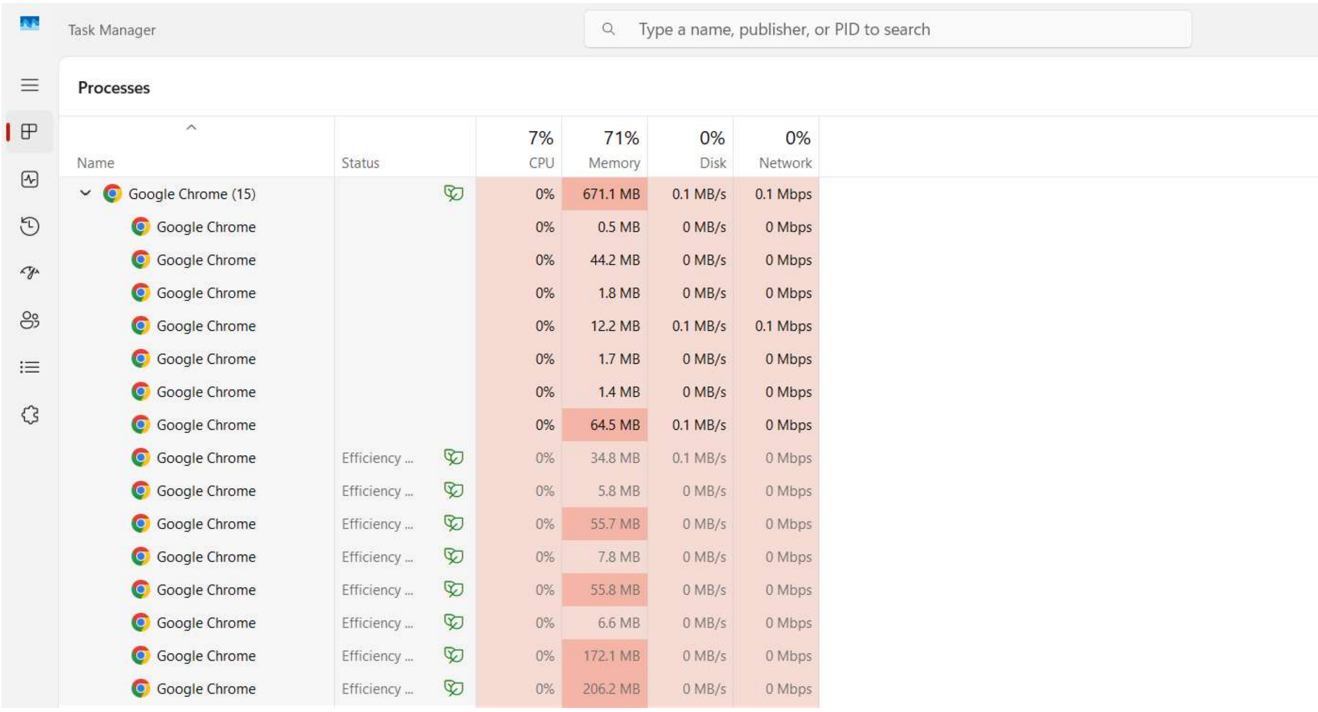
Domain Name System (response) Transaction ID: 0x0000 Flags: 0x8400 Standard query response, No error Questions: 0 Answer RRs: 2 Answers _googlecast._tcp.local: type PTR, class IN, cache flush: false PTR: Chromecast-Ultra._googlecast._tcp.local Additional Records Chromecast-Ultra._googlecast._tcp.local: type SRV, class IN Port: 8009 Target: 192-168-1-

DNS server. Devices send mDNS queries across the local subnet so local services (like `_googlecast` for Chromecast or `_companion-link` for Apple devices) can resolve host IP addresses dynamically.	45.local 192-168-1-45.local: type A, class IN, addr 192.168.1.45

2. Use the 'ping' command in Command Prompt or Terminal to test connectivity to www.spotify.com and www.zomato.com; record the response times and identify which site responds faster.

```
C:\Windows\System32>ping www.spotify.com

Pinging atc.spotify.map.fastly.net [2a04:4e42:31::810] with 32 bytes of data:
Reply from 2a04:4e42:31::810: time=151ms
Reply from 2a04:4e42:31::810: time=50ms
Reply from 2a04:4e42:31::810: time=147ms
Reply from 2a04:4e42:31::810: time=1139ms

Ping statistics for 2a04:4e42:31::810:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 50ms, Maximum = 1139ms, Average = 371ms

C:\Windows\System32>ping www.zomato.com

Pinging e70929.dsca.akamaiedge.net [2600:140f:1e00::1737:f4c3] with 32 bytes of data:
Reply from 2600:140f:1e00::1737:f4c3: time=48ms
Reply from 2600:140f:1e00::1737:f4c3: time=346ms
Reply from 2600:140f:1e00::1737:f4c3: time=258ms
Reply from 2600:140f:1e00::1737:f4c3: time=273ms

Ping statistics for 2600:140f:1e00::1737:f4c3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 48ms, Maximum = 346ms, Average = 231ms
```

3. Open Task Manager (Windows) or Activity Monitor (Mac) and list all running processes related to Chrome, WhatsApp Desktop, or any browser-based app you use; note the process name and memory usage for each.



Google Chrome (15 processes) — 671.1 MB total, 7% CPU, 0% Disk, 0% Network

Microsoft Edge (18)		1.5%	931.1 MB	0 MB/s	0.1 Mbps
Browser		0%	81.7 MB	0 MB/s	0 Mbps
Crashpad		0%	2.1 MB	0 MB/s	0 Mbps
GPU Process		0%	190.3 MB	0 MB/s	0 Mbps
Utility: Audio Service		0%	3.4 MB	0 MB/s	0 Mbps
Utility: Data Decoder Serv...		0%	3.1 MB	0 MB/s	0 Mbps
Utility: Network Service		0%	15.4 MB	0 MB/s	0.1 Mbps
Utility: Search Indexer Ser...		0%	3.4 MB	0 MB/s	0 Mbps
Utility: Storage Service		0%	4.5 MB	0 MB/s	0 Mbps
Utility: Wallet Service		0%	3.0 MB	0 MB/s	0 Mbps
Spare Renderer	Efficiency ...	0%	9.0 MB	0 MB/s	0 Mbps
Subframe: https://adnxs.c...	Efficiency ...	0%	18.2 MB	0 MB/s	0 Mbps
Subframe: https://adsdkpr...	Efficiency ...	0.4%	50.1 MB	0 MB/s	0 Mbps
Subframe: https://double...	Efficiency ...	0%	12.1 MB	0 MB/s	0 Mbps
Subframe: https://edgeco...	Efficiency ...	0%	31.3 MB	0 MB/s	0 Mbps
Tab: Facebook	Efficiency ...	0%	62.9 MB	0 MB/s	0 Mbps
Tab: Google	Efficiency ...	0%	36.6 MB	0 MB/s	0 Mbps
Tab: Maharashtra speaker ...	Efficiency ...	0.8%	157.2 MB	0 MB/s	0 Mbps
Tab: New tab	Efficiency ...	0.4%	246.8 MB	0 MB/s	0 Mbps

Microsoft Edge (18 processes) — 931.1 MB total, 1.5% CPU, 0.1 Mbps Network

4. Download and run the free version of 'Angry IP Scanner' or 'Advanced IP Scanner' to scan your local WiFi network; list the IP and device name of at least 2 devices connected to your network.

192.168.29.1 – Reliance Router

192.168.29.218 - Halo

5. Use ChatGPT or Bing AI to suggest 3 hardware tools used in network troubleshooting; for each, write one line describing when you would use it.

Here are 3 common hardware tools for network troubleshooting:

1. **Cable tester** — used to check if an Ethernet/RJ-45 cable has proper continuity and correctly wired pins (T568A/B), when you suspect a physical cable fault is causing no connectivity or intermittent link drops.

2. **Multimeter** — used to check voltage, continuity, and power supply health (e.g., PSU output, PoE injector voltage) when troubleshooting whether a networking device is failing due to a power issue rather than a network configuration issue.

3. **Toner probe (tone generator and probe)** — used to trace and identify a specific cable's physical path/endpoint in a messy cable bundle or patch panel, when you need to find which port a particular cable connects to without pulling apart the whole wiring.